Data Mining in Python

Today's goal • 2/3 items

Module 1
Basic Concepts of Data Mining

Welcome to the Course
- Welcome to Data Mining in Python — Video • 2 min
- MADSwPY Certificate Roadmap — Reading • 5 min
- Course Syllabus — Reading • 10 min
- Meet Your Fellow Learners — Discussion Prompt • 15 min
- Help Us Learn About You — Reading • 10 min

Lesson 1: Introduction to Data Mining and Its Basic Functionalities
- What is Data Mining — Video • 14 min
- Introduction to the Basic Functionalities of Data Mining — Reading • 10 min
- Data Mining Functionalities (Part 1) — Video • 6 min
- Data Mining Functionalities (Part 2) — Video • 7 min
- Data Mining Functionalities (Part 3) — Video • 4 min
- Knowledge Check: Basic Functionalities of Data Mining — Practice Assignment • 15 min

Lesson 2: Representations of Real World Data
- Introduction to Basic Data Representations — Reading • 10 min
- Representing Itemsets, Vectors, and Matrices — Video • 6 min
- Representing Sequences — Video • 4 min
- Knowledge Check: Basic Data Representations (Part 1) — Practice Assignment • 15 min
- Representing Time-Series and Spatial/Temporal Data — Video • 8 min
- Representing Graph Data — Video • 5 min
- Representing Stream Data — Video • 4 min
- Case Study: Representations of Real-World Text Data — Reading • 10 min
- Knowledge Check: Basic Data Representations (Part 2) — Practice Assignment • 15 min

Lesson 3: Patterns and Similarities
- Introduction to Patterns and Similarities — Reading • 10 min
- Data Mining Based on Patterns — Video • 5 min
- Data Mining Based on Similarities — Video • 6 min

Module 1 Wrap Up
- Module 1 Quiz: Introduction to Data Mining — Graded Assignment • 20 min
- Introduction to Module 1 Programming Assignment: Visualizing Different Data — Reading • 10 min
- Module 1 Programming Assignment: Warming Up — Programming Assignment • 3h
- Dialogue Reflection — Dialogue • 15 min
- Module 1 Optional Readings & Resources — Reading • 10 min

Module 2
Mining Itemset Data

Module 3
Mining Vector and Matrix Data

Module 4
Mining Sequences

Course Syllabus

0:31 / 12:11 2x

Welcome

Welcome to the *Introduction to Data Mining* course! This is the first course in the ***More Applied Data Science with Python (MADSwPy) Specialization***. An essential prerequisite for this Specialization is successfully completing the ***Applied Data with Python Specialization*** on Coursera.

Data mining is concerned with discovering useful knowledge from large-scale datasets. This course will introduce you to the basic concepts and general data mining tasks. It focuses on formally describing real-world information with general data representations (e.g., itemsets, vectors, matrices, sequences, time series, and networks) that facilitate downstream analytics tasks. You will learn how to formulate data in the wild with one or more of these representations and characterize each type of data through pattern extraction and similarity measures, which are basic building blocks for advanced analysis and machine learning models. The course then delves into an overview of four basic representations of data:

- Itemset data;
- Vector data;
- Matrix data;
- Sequence data;

You will learn the formal definitions of these data representations, corresponding real-world examples, and methods for extracting patterns and computing similarities from each particular type of data. By successfully completing this course, you will be able to:

- Comprehend the basic concepts and general tasks of data mining
- Describe real-world information with general data representations
- Recognize general data representations' role in facilitating downstream analytics tasks
- Formulate data in the wild with one or more types of data representations
- Characterize each type of data using methods of pattern extraction and similarity measures
- Develop foundational capacity to learn advanced analysis and machine learning models

You won't need a textbook for this specialization or this course—all of the concepts, principles, procedures, and facts are described in the video lessons and readings, along with using Coursera to run Python for practice activities and graded assessments.

Your instructor for this course is Dr. Qiaozhu Mei, a Professor of Information at the University of Michigan School of Information and a Professor of Electrical Engineering and Computer Science at the University of Michigan College of Engineering. Dr. Mei is a scholar with a broad interest in real-world problems related to text information management. His research focuses on information retrieval and text mining, with applications in the web, social media, scientific literature, bioinformatics, and health informatics. He also has a strong interest in machine learning, data mining, natural language processing, and social network analysis. To know more about Dr. Qiaozhu Mei's research, please see his personal website.

Participation Strategies

Engaged learning looks different for everybody. In this course, we hope you will define your own measures of success and engage with the material in a way that best suits your needs.We recognize and celebrate the diverse ways learners engage in courses. As you go through this course, we hope you will reflect on your unique skills, needs, and aspirations, and engage in the course material in a way that aligns with your own goals. While the course provides time estimates for completion, you should feel empowered to engage in the material in whatever ways make sense to you.

Course Structure & Grading Policy

The course is structured into four modules. In each module, we have included lecture videos, ungraded knowledge check assignments, auto-graded assignments, and programming assignments.

Below, you will find a short description of each Module's topics. Past that, you will find information on passing thresholds and the weights assigned to each assignment type that will make up your course grade.

Module 1: Basic Concepts of Data Mining

- Introduction to Data Mining
- Basic Functionalities of Data Mining
- Representations of Real World Data
- Patterns and Similarities

Module 2: Mining Itemset Data

- Mining Frequent Itemsets
- Evaluating Frequent Itemsets
- Similarity of Itemsets

Module 3: Mining Vector and Matrix Data

- Vector Representation of Data
- Similarity of Vectors
- Patterns in Matrix Data

Module 4: Mining Sequences

- Sequence Representation of Data
- Sequential Patterns
- Sequence Similarity
- Moving Forward

Assessments	Weight
Module 1 Quiz: Introduction to Data Mining	5%
Module 1 Programming Assignment: Visualizing Different Data	5%
Module 2 Quiz: Mining Itemset Data	5%
Module 2 Programming Assignment: Dealing with Itemset Real-World Data	25%
Module 3 Quiz: Mining Vector and Matrix Data	5%
Module 3 Programming Assignment: Dealing with Vector and Matrix Real-World Data	25%
Module 4 Quiz: Mining Vector and Matrix Data	5%
Module 4 Programming Assignment: Dealing with Sequences Real-World Data	25%
Total	**100%**

Community Engagement

We expect everyone to be mindful of what they say and its potential impact on others. The goal is to have respectful discussions that do not violate the community space created for these conversations. Here are some productive ways to engage in this course:

- **Participate:** This is a community. Read what others have written and share your thoughts.
- **Stay curious:** Learn from experts and each other by listening and asking questions, not making assumptions.
- **Keep your passion positive:** When replying to a discussion forum post, respond with thoughts on what was said, not about the person who posted. Avoid using all caps, too many exclamation points, or aggressive language.
- **Acknowledge discomfort:** The topics discussed in this course might be challenging or hard to talk about. Stick with it and remember the benefits of having these tough conversations that surface from multiple perspectives.

We expect all learners to abide by our full Learner Engagement Policy. We will specifically be monitoring this course for language that could be considered inflammatory, incivil, racist, or otherwise unacceptable for this learning space, and we will remove language deemed such.

Note regarding "study groups" outside of the platform: While learners are encouraged to interact with one another using communication tools offered within the learning platform, note that any study groups formed outside of this platform, such as over social media or communication applications and websites (e.g., WhatsApp) are not affiliated, endorsed, or moderated by the University of Michigan. If you receive an invitation to an outside study group mentioning the course and claiming to have any official connection to U-M or individual instructors, please exercise caution as this may be an attempted scam.

Academic Honesty

All submitted work should be your own and academic dishonesty is not allowed, such as, but not limited to:

- Copying words, ideas, or other materials from another source without giving credit to the original author
- Copying answers or copying from your peers within the course
- Employing or allowing another person to alter or revise your work, and then submitting the work as your own
- Using Artificial Intelligence tools, such as ChatGPT, to create or edit your work and submitting that work as your own, unless you were instructed to use AI as part of the assignment

Academic dishonesty may result in deleting assignment submissions, forum responses, removal from the course, or other actions as needed. For more information on academic honesty and misconduct, please refer to our Learner Engagement Policy.

Course Support

Questions and discussion of course material should take place within the course itself. Please do not contact instructors or teaching assistants off the platform, as responding to individual questions is virtually impossible.

We encourage you to direct your questions to **Course Support Forum** where your question might be answered by a fellow learner or one of our course team members. For technical help please contact the Coursera Learner Help Center support forums.

Accessibility

We are committed to developing accessible learning experiences for the widest possible audience. We recognize that learners with disabilities (including but not limited to visual impairments, hearing impairments, cognitive disabilities, or motor disabilities) might need more specific accessibility-related support to achieve learning goals in this course.

Please use the accessibility feedback form to let us know about any accessibility challenges such as urgent issues that keep you from making progress in the course (e.g., missing or inadequate alt-text, captioning errors).

Third-party sites and software: While the University of Michigan is not responsible for the accessibility of third-party sites or applications that may be linked from this course, we still encourage you to report associated third-party accessibility issues so that we can ensure you are able to participate. In such cases, we may contact you for additional information as we investigate ways of removing accessibility barriers or to suggest accessible alternatives.

Global Learning Statement

We welcome all learners to this course. By enrolling in this course, you join people from all over the world and we value the perspectives they bring to the learning environment. We strive to create a community of mutual respect and trust where people from all backgrounds and views are valued and heard without the threat of bias, harassment, intimidation, or discrimination. We pay attention to your feedback on how different types of learners experience this course and aim to make improvements so the course can best serve everyone. We hope you enjoy learning about topics that are important to you.

Tools & Technology

Coursera Platform

If you are new to Coursera and you want to familiarize yourself with the platform, read the Help Articles in Coursera Help Center. This is a good resource where you can find answers to many basic questions such as how to adjust video settings, how to submit assignments, and how to earn a course certificate.

H5P

This course includes course content and learning activities built with a tool called H5P to include additional interactivity in this course. These activities will appear within the course itself, to avoid redirecting you outside of the learning platform. If you encounter challenges when exploring the H5P interactive elements, we recommend that you check the course discussion forums on Technical Support for assistance.

GenAI Tools

Please note that any independent use of GenAI tools to complete assignments remains subject to the Academic Honesty section of this syllabus. The use of these tools can also introduce unique copyright, privacy, and ethical considerations. Please review the Center for Academic Innovation's Generative AI Guidance for Learners to learn more.

Coursera Coach

During the course you may encounter Coursera Coach, a generative AI-assisted tool developed by Coursera and supported on the Coursera platform. Using the Coursera Coach is optional. Please note that you should verify AI-generated content for accuracy, The Coursera Coach should not replace your personal analysis of the content.

Jupyter Notebooks

In this course, the Jupyter Notebook tool is used to provide learners a programming environment to work in, directly on the Coursera platform through Coursera Labs. This environment utilizes Jupyter Notebooks to combine code and text.

This environment is intended to provide a ready-made workspace for learners to interact with course materials and without the need for local setup. These notebooks are available for learners to run locally, however support can only be provided for learners who utilize the Coursera Labs environment.

Mark as completed

Like Dislike Report an issue

Go to next item